

Software Requirements Specification (SRS)

Hospital Management System

Project: Hospital Management System

Version: 1.0

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Revision history

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1.0	4th September 2026	Aniket Raosaheb Jadhav, Aniket S Patil, Anirudh Nagarajan, Anirudha Alagawadi	Initial draft SRS for Hospital Management System	Yes , approved

Approvals

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1. Introduction

1.1 Purpose

This document is a Software Requirements Specification (SRS) for a Hospital Management System (HMS). It defines the functional and non-functional requirements, external interfaces, and verification criteria for a patient record and appointment management system for hospitals, intended for use by developers, evaluators, and instructors.

1.2 Scope

The system covers patient registration, appointment scheduling, electronic patient records (medical history, diagnoses, prescriptions), billing and invoicing, staff/doctor management, and operational reporting. The system will be developed in C / C++ as a desktop / client-server application backed by a relational database. It excludes deep integration with external insurance clearing houses and national health information exchanges, which are out of scope for this version.

1.3 Audience

Developers, QA Engineers, Hospital Administrators, Doctors, Receptionists, and Assessment Evaluators.

1.4 Definitions

List of acronyms: HMS (Hospital Management System), EHR (Electronic Health Record), OPD (Out-Patient Department), IPD (In-Patient Department), PHI (Protected Health Information), PII (Personally Identifiable Information), UI (User Interface), API (Application Programming Interface), DB (Database), RBAC (Role-Based Access Control), RTO/RPO (Recovery Time/Point Objective).

2. Overall description

2.1 Product perspective

The HMS is a standalone / client-server application used at hospital front-desk, consultation, and billing counters. It is implemented in C / C++ and interacts with a relational database (e.g., MySQL / SQLite) for persistent storage of patient, appointment, billing, and inventory data. It focuses on core patient record and appointment management functionality without dependency on peripheral hardware or external notification gateways.

2.2 Major product functions (detailed)

- User authentication and role-based access control
- Patient registration and demographic management
- Electronic patient record management (history, diagnosis, prescriptions)
- Appointment scheduling, rescheduling, and cancellation
- Doctor and staff profile and schedule management
- Billing, invoicing, and payment recording
- Operational and financial report generation
- Audit logging of sensitive data access and modification

2.3 User roles and characteristics (expanded)

- Patient: basic computer literacy; interacts mainly through the receptionist or a simple self-service kiosk; expects short waiting/registration times.
- Receptionist: front-desk staff; performs registration and appointment booking; needs a fast, simple UI.
- Doctor: reviews and updates patient records and prescriptions; needs quick access to patient history during consultation.

- Administrator: manages staff accounts, schedules, system configuration, and generates reports.

2.4 Operating environment

Desktop / client-server application running on Windows or Linux workstations at hospital counters, connected over a local area network (LAN) to a central database server. Suitable UPS backup is assumed at the server to protect against power interruptions.

2.5 Constraints

Compliance with applicable healthcare data protection regulations (e.g., HIPAA-equivalent local regulation) for PHI/PII; implementation constrained to C / C++ with a relational database backend; hardware constraints of counter workstations; limited network bandwidth in some deployment sites.

3. External interface requirements

3.1 User interfaces

Primary UI: desktop GUI (or structured console-based menu) with clear, role-specific menu structures for Receptionist, Doctor, and Administrator. Accessibility: high-contrast mode and keyboard-navigable forms for staff with visual constraints.

3.2 Software interfaces

- Relational Database Interface (SQL) for persistent storage of patients, appointments, billing, and inventory data.
- Reporting module for generating operational and financial reports.

3.3 Communications

- TLS 1.2+ enforced for any networked client-server communication.
- Retry / offline queuing strategy for transactions when the central database is temporarily unreachable from a satellite counter.

[Overall project provides at least 15 Functional Requirements, 5 Non-Functional Requirements, 4 Security Objectives, and 5 Security Requirements, as detailed in Sections 4 and 5.]

4. System features (detailed)

Each requirement below includes acceptance criteria and a reference test case. IDs follow HMS-F-###.

4.1 Authentication & Access Control

Description: Authenticate staff users and enforce role-based access to system modules.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
HMS-F-001	The system shall authenticate users via a unique username and password before granting access to any module.	Functional	High	Security	AC-HMS-F-001: Invalid credentials are rejected; valid credentials grant a session. Test: TC-Auth-01	Requires user account store
HMS-F-002	The system shall enforce role-based access control (Admin, Doctor, Receptionist, Billing Staff).	Functional	High	Security / Admin	AC-HMS-F-002: Users can only access screens/functions permitted for their role. Test: TC-Auth-02	Depends on HMS-F-001

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
HMS-F-003	The system shall lock a user account after 5 consecutive failed login attempts and record an audit event.	Functional	High	Security	AC-HMS-F-003: 6th failed attempt blocks login and logs event. Test: TC-Auth-03	Admin unlock required

4.2 Patient Registration & Records

Description: Register patients and maintain accurate, auditable electronic health records.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
HMS-F-010	The system shall allow a receptionist to register a new patient with demographic and contact details, generating a unique Patient ID.	Functional	High	Receptionist	AC-HMS-F-010: New patient saved with unique ID; duplicate IDs rejected. Test: TC-Reg-01	
HMS-F-011	The system shall allow authorized staff to search patient records by Patient ID, name, or phone number.	Functional	High	Receptionist / Doctor	AC-HMS-F-011: Search returns matching record(s) within 2s. Test: TC-Reg-02	
HMS-F-012	The system shall allow doctors to view and update a patient's medical history, diagnoses, and prescriptions.	Functional	High	Doctor	AC-HMS-F-012: Updated entry visible immediately and versioned. Test: TC-Rec-01	Requires HMS-F-002 (Doctor role)
HMS-F-013	The system shall maintain a complete audit trail of all changes made to a patient record, including user, timestamp, and action.	Functional	High	Compliance	AC-HMS-F-013: Every create/update/delete on a record produces an audit entry. Test: TC-Rec-02	Supports audit and compliance reporting

4.3 Appointment Scheduling

Description: Book, reschedule, and cancel appointments while avoiding scheduling conflicts.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
HMS-F-020	The system shall allow a patient or receptionist to book an appointment with a specific doctor for an available time slot.	Functional	High	Patient / Receptionist	AC-HMS-F-020: Booking succeeds only for a free slot; confirmation shown. Test: TC-Appt-01	
HMS-F-021	The system shall prevent double-booking of the same doctor time slot.	Functional	High	Operations	AC-HMS-F-021: Concurrent booking attempts for same slot: only one succeeds. Test: TC-Appt-02	Requires transaction locking
HMS-F-022	The system shall allow cancellation or rescheduling of an existing appointment, updating the doctor's calendar in real time.	Functional	Medium	Patient / Receptionist	AC-HMS-F-022: Slot released or moved immediately on confirmation. Test: TC-Appt-03	

4.4 Billing & Invoicing

Description: Generate accurate invoices and record payments against patient visits.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
HMS-F-030	The system shall generate an itemized invoice for consultation and tests at the end of a visit.	Functional	High	Billing Staff	AC-HMS-F-030: Invoice total equals sum of line items. Test: TC-Bill-01	
HMS-F-031	The system shall record payments (cash / card / insurance) against an invoice and update the outstanding balance.	Functional	High	Billing Staff	AC-HMS-F-031: Balance decreases by payment amount; overpayment rejected. Test: TC-Bill-02	

4.5 Doctor / Staff Management

Description: Manage staff schedules, prescriptions, and operational reporting.

Req ID	Requirement (shall...)	Type	Priority	Source/Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
HMS-F-040	The system shall allow an administrator to manage doctor/staff profiles, specialties, and working schedules.	Functional	Medium	Administrator	AC-HMS-F-040: Profile changes reflected in appointment availability. Test: TC-Staff-01	
HMS-F-042	The system shall allow a doctor to issue an electronic prescription linked to the patient record.	Functional	High	Doctor	AC-HMS-F-042: Prescription attached to the patient record and available for review. Test: TC-Rx-01	Depends on HMS-F-012
HMS-F-043	The system shall allow a doctor to generate and export a patient visit summary (diagnosis, prescriptions, and follow-up instructions) at the end of a consultation.	Functional	Medium	Doctor	AC-HMS-F-043: Visit summary contains diagnosis, prescriptions, and follow-up notes, and matches the underlying record. Test: TC-Doc-01	Depends on HMS-F-012

5. Non-functional requirements (detailed)

NFRs below are measurable and tied to test plans. IDs HMS-NF-####.

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
HMS-NF-001	Appointment search and booking response time shall be ≤ 2 seconds for 95% of transactions under normal load.	Performance	High	95th percentile ≤ 2 s in production-like test. Test: TC-Perf-01
HMS-NF-002	The system shall provide high availability during hospital operating hours, with scheduled maintenance windows excluded.	Reliability	High	Uptime reports show consistently high availability per month. Test: Ops reports.
HMS-NF-003	All patient health data (PHI/PII) must comply with applicable data-protection regulations; sensitive data must not be stored in plaintext.	Security / Compliance	High	Compliance audit checklist pass. Test: TC-Sec-01
HMS-NF-004	The system shall provide audit logs with timestamped events retained for a minimum of 7 years for audit purposes.	Audit / Data Retention	High	Automated archival and retrieval verified. Test: TC-Ops-01
HMS-NF-005	The system's user interface shall be operable by staff with no more than 30 minutes of training, measured via a	Usability	Medium	Standardized usability test: new users trained for exactly 30 minutes, then attempt core tasks unaided. $\geq 90\%$ complete registration,

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
	standardized usability test on core workflows (registration, booking, billing).			booking, and billing tasks without assistance. Test: TC-Usab-01

5.1. Security

5.1.1 Security Objectives

- SO-1: Protect patient confidentiality (PHI/PII) from unauthorized access, disclosure, or modification.
- SO-2: Ensure the integrity and availability of medical records and appointment data.
- SO-3: Provide accountability through auditable logs of all access to sensitive data.
- SO-4: Prevent unauthorized privilege escalation between user roles.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case ref
HMS-SR-001	TLS 1.2+ shall be mandatory for all networked client-server connections.	Security	High	AC-SR-001: Connection attempts below TLS 1.2 are refused. Test: TC-Sec-02
HMS-SR-002	User passwords shall be stored using a salted cryptographic hash and never stored or logged in plaintext.	Security	High	AC-SR-002: Database inspection shows no plaintext passwords. Test: TC-Sec-03
HMS-SR-003	Role-based access control shall be enforced at both the UI layer and the data-access layer.	Security	High	AC-SR-003: Direct data-layer requests outside a user's role are rejected. Test: TC-Sec-04
HMS-SR-004	User sessions shall automatically terminate after 15 minutes of inactivity, requiring re-authentication before further access.	Security	High	AC-SR-004: Session is invalidated and user is redirected to login after 15 minutes idle; no data access possible without re-authentication. Test: TC-Sec-05
HMS-SR-005	Sensitive patient identifiers stored in the database (e.g., national ID equivalents) shall be protected using a one-way cryptographic hash rather than stored in plaintext, as a basic at-rest safeguard.	Security	Medium	AC-SR-005: Database inspection shows sensitive identifier fields stored as hashes, not plaintext. Test: TC-Sec-06

6. Quality attributes & Acceptance tests

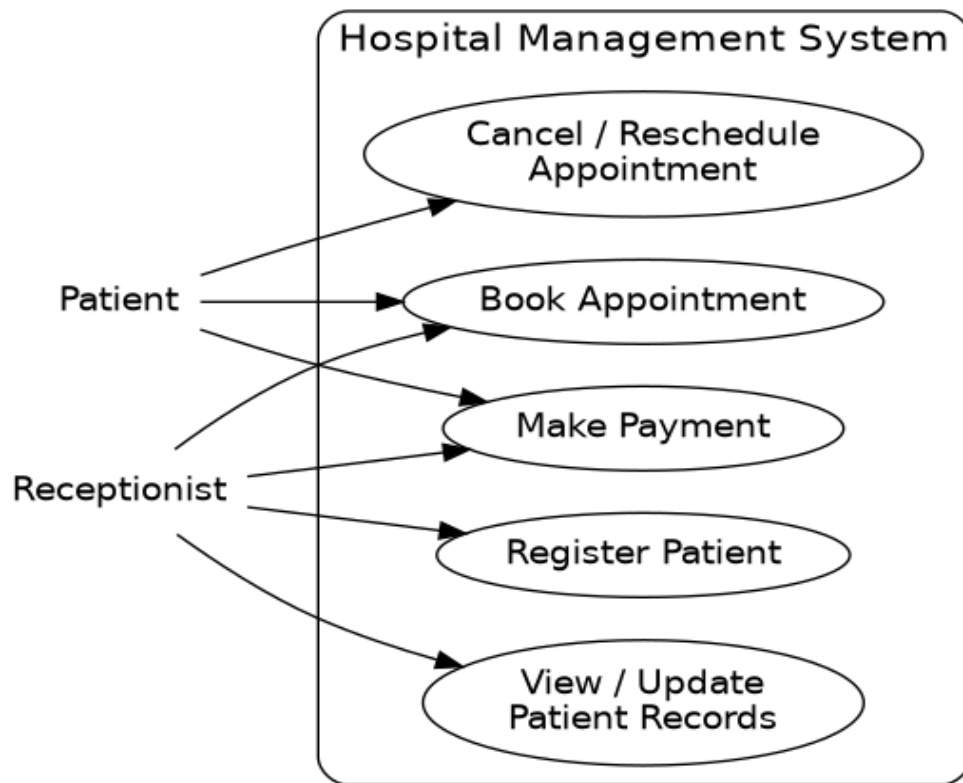
Exit criteria for acceptance: All high-priority functional requirements implemented and verified, no critical NFR failures, and the RTM shows all test cases passed.

Acceptance test suites: Authentication, Patient Registration, Appointment Scheduling, Billing, Reporting, Performance, and Security tests.

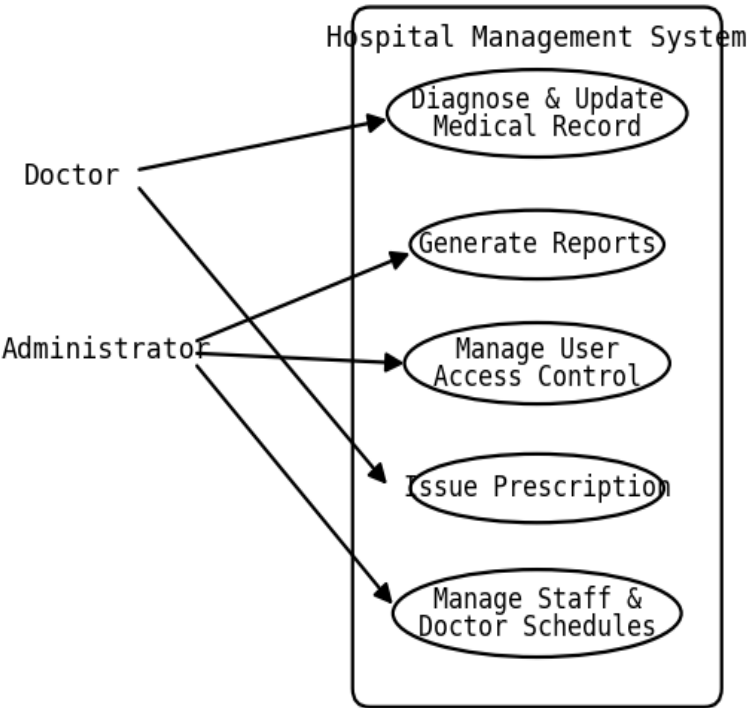
7. System models and diagrams

7.1 UML Use-Case diagrams

Use-case diagram 1: Patient and Receptionist interactions.



Use-case diagram 2: Doctor and Administrator interactions.



8. Requirements Traceability Matrix (RTM)

Req ID	Requirement (short)	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
HMS-F-001	Authenticate user	4.1 / DS-Auth-01	AuthModule	TC-Auth-01	N	
HMS-F-002	Role-based access	4.1 / DS-Auth-02	AuthModule	TC-Auth-02	N	
HMS-F-010	Register patient	4.2 / DS-Reg-01	PatientModule	TC-Reg-01	N	
HMS-F-012	Update medical record	4.2 / DS-Rec-01	PatientModule	TC-Rec-01	N	
HMS-F-020	Book appointment	4.3 / DS-Appt-01	AppointmentModule	TC-Appt-01	N	
HMS-F-021	Prevent double-booking	4.3 / DS-Appt-02	AppointmentModule	TC-Appt-02	N	
HMS-F-030	Generate invoice	4.4 / DS-Bill-01	BillingModule	TC-Bill-01	N	
HMS-F-042	Issue prescription	4.5 / DS-Pharm-01	PrescriptionModule	TC-Rx-01	N	
HMS-NF-001	Response time target	5 / DS-Perf-01	AppointmentModule	TC-Perf-01	N	
HMS-SR-002	Password hashing	5.1.2 / DS-Sec-01	AuthModule	TC-Sec-03	N	
HMS-F-043	Generate visit summary	4.5 / DS-Doc-01	PatientModule	TC-Doc-01	N	
HMS-NF-005	Usability / training time	5 / DS-Usab-01	UIModule	TC-Usab-01	N	
HMS-SR-004	Session timeout	5.1.2 / DS-Sec-02	AuthModule	TC-Sec-05	N	
HMS-SR-005	Sensitive ID hashing	5.1.2 / DS-Sec-03	PatientModule	TC-Sec-06	N	

Status legend: N = Not tested, P = Pass, A = Fail (needs Action).